30/8/26 EXP NO: 7

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

Aim

To write a program to implement flow control at the data link layer.

Algorithm

- start program
- read total no. of frames
- initialize base = 0 & next step = 0
- repeat while base < N
- stop program

Program

```
def simulate (total frames, window-size, last-frame):
    base 0
    next_seq = 0
    while base < total_frams
        window_end = min (base+ window_size 1,
                          total frames_1)
    print ("In:..")
    print (f" sender window: (fram {base} - fram
                          {window_end})' )
    while next_seq < base + window_size and
    while next_seq < base + window_size and
```

```
next_seq < total_frames:
print(f" Node A -> sending Frame {next_seq}")
if next.seq == cost_frame:
    print(f"(Frame {next_seq} lost in transit!)")
    next_seq+=1
if lost_frame is not none and base
    <= lost_frame < next_seq:
print(f" Node B --> Frame {base_frame} <
    next_seq:
    print(f" Node B--> Frame {base_frame}
Missing Discarding out_of_order frams
    after if.)
print(f" Node A --> sending Frame {next_seq}")
if next_seq == lost_frame:
    print(f" (frame {next_seq} lost in transit!)")
next_seq += 1
if lost_frame is not Non and base <=
    lost frame < next_seq:
print(f " Node B -> frame {lost_frame}
    missing discarding out of order frames
    after it.")
```

Output:

enter tot no. of frame : 8

senders window size : 4

sender window: [frame 0 - fame 3]

Node A → sending fame 0

Node A → sending fame 1

Node A → sending Frame 2

.
.
.

Node A → Ack received

sender window

Node A → sending frame 4

.
.
.

Node A → timeout

```
print (f" Node A --> Timeout | Resending window
    starting from Frame {last_frame}").
next_seq = lost_frame}")
next_seq = lost_frame
lost_frame = None
continue
for i in range (base, next_seq):
    print (f" Node B --> Received frame{i},
        sending Ack {i+1}")
    print (f" Node A --> Ack {next_seq-1}
        received sliding window forward.")
    base = next_seq
    print ("\n...")
Print (f" All {total_frames} frames
    successfully transmitted from Node A
    to Node B)

def main ():
    print ("... SLIDING WINDOW PROTOCOL_FLOW
        CONTROL SIMULATION (GO BACK_N) ==\n")
total_frames = int (input ("Enter total no. of frames:"))
window_size = int (input ("Enter sender
        window size:"))
```

```
    if lost_frame == -1:
        lost_frame = None
        simulate(total_frames, window_size,
                 lost_frame)

if __name__ == "__main__":
    main()
```

}

Result

The ~~~~ code was written and the program was implemented successfully.

18/2/24